

Terry Liu 劉冠廷 | Portfolio Summary

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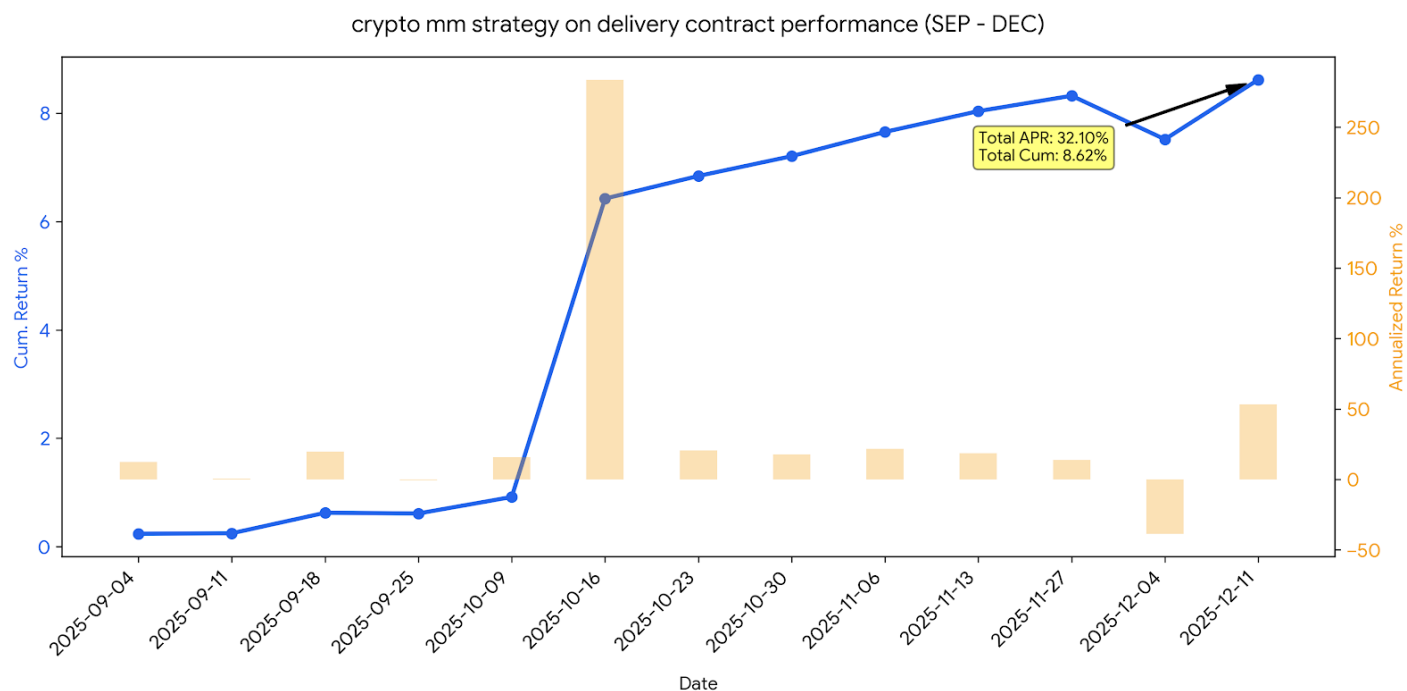
 GitHub: <https://github.com/kiwicabbage> (https://github.com/kiwicabbage)

1. Cryptocurrency Market Making Strategy for Low-Liquidity Contracts

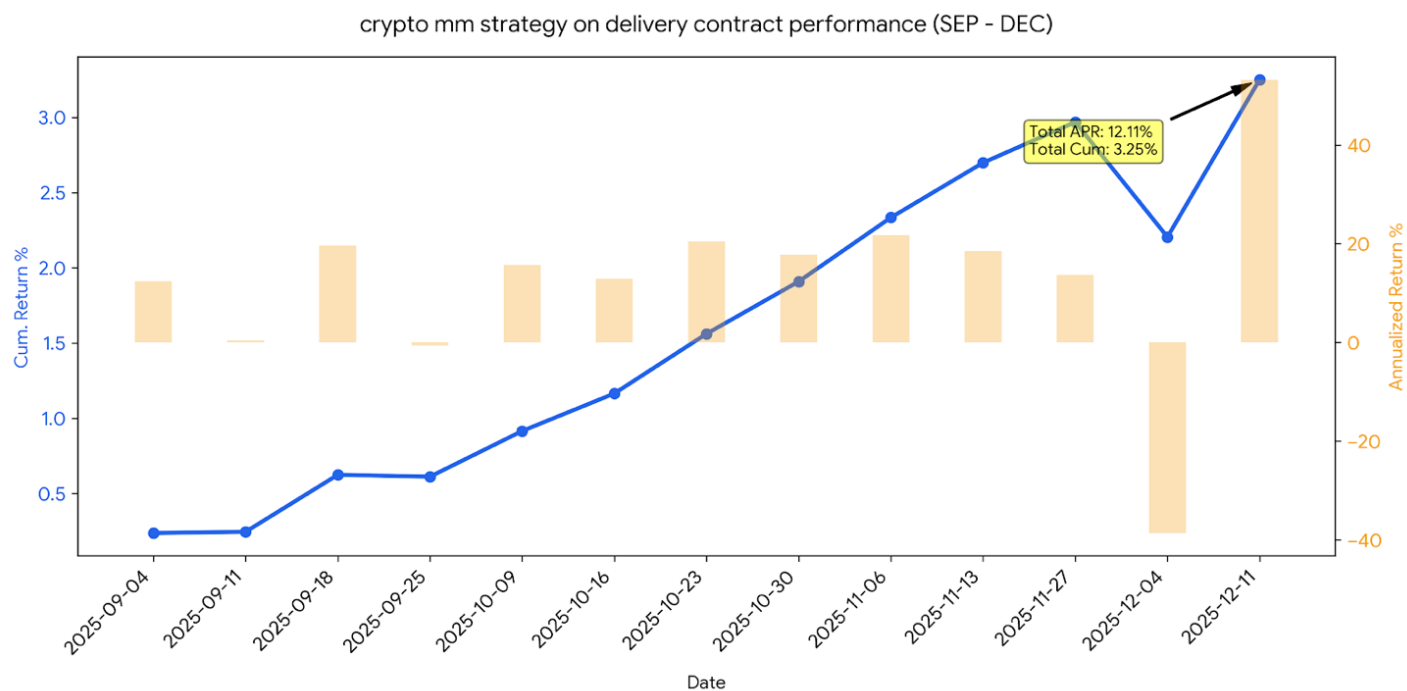
A crypto arbitrage strategy that provides liquidity through market making in low-volume markets, with a specialized focus on delivery futures.

This strategy primarily focuses on market making for delivery futures contracts on Bybit. It leverages a predictive model to determine the optimal quoting prices and select the best hedging instruments, while dynamically factoring in transaction costs and funding rate fluctuations.

- **Real performance during 2025**



- Real performance without outliers



2. Integrating Noise-Reduction and ML Models for Enhanced Intraday Trading

<https://github.com/kiwicabbage/Thesis-Integrating-Noise-Reduction-and-ML-Models>

(<https://github.com/kiwicabbage/Thesis-Integrating-Noise-Reduction-and-ML-Models>)

Predicts the direction of index future price of Taiwan and HK market, combined the ML model with noise-reduction technique.

- The best results were obtained from the combination of expanding window updates with denoised targets, achieving the most stable and profitable performance.

This project develops a machine learning-based CTA (Commodity Trading Advisor) strategy using Classification Model to predict directional movements of Taiwan Index Futures. The study evaluates different factor combinations and classification targets, along with three model update mechanisms: fixed parameters, rolling window updates, and expanding window updates.

To enhance signal stability, a denoising process was applied to the classification labels, aiming to reduce the influence of high-frequency noise. Key findings include:

- Expanding window updates significantly improved strategy performance.
- Using denoised targets helped reduce overfitting and enhanced generalization.

This strategy development framework is flexible and scalable, making it applicable to multi-asset and cross-market CTA strategies.

- **Back test result (all out sample using expanding window to train the model)**



3. Taiwan Stock Market Open Price Prediction via Pre-Market Limit Order Book Data

<https://github.com/kiwicabbage/LOBster-NN> (<https://github.com/kiwicabbage/LOBster-NN>)

Predicts the opening price of Taiwan 0050 ETF constituent stocks by feeding pre-market Limit Order Book into deep learning models — CNN, MLP, and DeepLOB — before the call auction closes at 09:00.

- The best performance was achieved by the **MLP model (v2)**, which reduced MSE loss from **239,135** → **25,504** over 40 epochs — a **~89% reduction** — demonstrating strong convergence on the pre-market signal.

This project develops a deep learning pipeline to predict the opening price of TWSE 0050 ETF constituent stocks using pre-market Limit Order Book (LOB) tick data. The Taiwan Stock Exchange runs a call auction from 08:30 to 09:00 before the market opens, during which limit orders accumulate in the order book but no trades are executed. This project exploits the information embedded in those pre-open LOB snapshots — bid/ask prices and volumes at 5 depth levels — to predict the opening quote before the auction closes.

The dataset consists of tick-level LOB data for 50 stocks across 13 trading days in June 2024, yielding approximately 650 (day × stock) training samples. A key technical challenge was correctly parsing the TWSE data feed's status codes: separating `status=128` (pre-market snapshot) from `status=8` (opening tick), which is a non-obvious but critical preprocessing step.

Three model architectures were explored across successive iterations:

- **v1 — Simple CNN Regressor:** Treats the LOB sequence as a 2-D image and applies stacked Conv2D layers with adaptive average pooling. A Softplus activation enforces strictly positive price outputs. Trained for 20 epochs.
- **v2 — MLP Regressor:** Flattens the LOB snapshot sequence into a feature vector and passes it through a 3-layer fully-connected network, enabling faster pipeline iteration. Achieved best loss of **25,504 MSE at epoch 40**.
- **v3 — DeepLOB (in progress):** Adapts the architecture from Zhang et al. (IEEE TSP, 2019) — combining three CNN feature extraction blocks, a multi-scale Inception module, and an LSTM — re-targeted for the Taiwan pre-market setting.

Key findings include:

- **Pre-market LOB order flow contains predictive signal for the opening quote,** even with a small dataset of 13 trading days.
- **Iterating from CNN (v1) to MLP (v2) revealed that simpler architectures can outperform when data volume is limited,** avoiding the risk of overfitting to spatial patterns in short sequences.
- **The DeepLOB architecture (v3) is designed to capture both depth-of-book**

structure and temporal dynamics simultaneously, and represents the natural next step for scaling this research to longer data horizons.

This research framework is directly applicable to pre-open execution strategies, opening price discovery analysis, and cross-stock volatility prediction on the TWSE.

4. Automated Options Short Straddle Trading System

<https://github.com/kiwicabbage/sj-point> (<https://github.com/kiwicabbage/sj-point>)

Using Statistical Model predict the best spot to place an short straddle strategy on Taiwan Option Market

This project involves building an automated short straddle trading system for Taiwan Option Market.

The system integrates data acquisition, risk control, trading signal generation, and order execution modules, all organized in a modular and maintainable architecture. A real-time monitoring interface was developed using PyQt6 to ensure transparency and stability during live trading.

5. Stable Coin Arbitrage Strategy

<https://github.com/kiwicabbage/bn-bfUSD-arb> (<https://github.com/kiwicabbage/bn-bfUSD-arb>)

High Frequency Arbitrage while there is a premium on BFUSD

6. LOB Data Collector

<https://github.com/kiwicabbage/HFT-crypto-data-fetch> (<https://github.com/kiwicabbage/HFT-crypto-data-fetch>)

Collect high frequency limit order book (LOB) data and reorganized to GCS

7. DeFi Arbitrage Dashboard

<https://github.com/kiwicabbage/Hyperliquid-FundingRate-Dashboard>

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A dash board aim to check the arbitrage opportunities